

CMS Open Data H to tau tau Search: Phase 4c Audit-Corrected Observed Results

Analysis my_analysis

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Change Log

Phase 4c v2 audit correction

- Added a same-sign data-driven QCD/fake background estimate.
- Added multivariate data/MC input reweighting derived in validation/control regions before classifier training.
- Promoted the calibrated categorized score model to the primary observed result.
- Widened DY/Z normalization to reflect the absence of embedding and EWK Z reduced samples.

1 Phase 4c Observed Inference: Audit-Corrected Full Data

1.1 Summary

Phase 4c has been updated after the audit of the large expected versus observed discrepancy. The problem was physics modelling: the original score-template fit omitted a reducible QCD/fake-tau component, used background MC before correcting large data/MC input-shape differences, and carried too restrictive a DY/Z normalization model for a reduced sample without embedding or EWK Z simulation.

The primary observed result is therefore the calibrated trained-discriminator fit in the same VBF, boosted, and zero-jet categories. The classifier is trained only on MC, but the background MC receives a multivariate data/MC input reweighting derived before training in W high-mT, same-sign, Z-rich, and top/b-tag validation/control regions. The primary score model also includes the same-sign data-driven QCD/fake template and wider DY/Z normalization nuisance terms. No alternative mass-fit result is quoted in this Phase 4c note; the frozen previous visible-mass baseline is preserved only for the final signal-strength comparison.

1.2 W and QCD Control Inputs

The full-data W+jets scale from the high-mT control region is 0.8528 ± 0.0370 . The VBF background control scale from the VBF-like top-btag non-signal region is 0.5571 ± 0.0515 ; it is applied only to MC backgrounds in the VBF fit category. This addresses the large VBF data/MC overprediction seen in the original observed templates without changing the signal normalization or the boosted/zero-jet categories. The same-sign QCD/fake estimate uses data

minus non-QCD MC in the same-sign low-mT region and a transfer factor measured in the lowest fitted-observable bin. For the primary calibrated-score model this factor is 0.0204 ± 0.2533 .

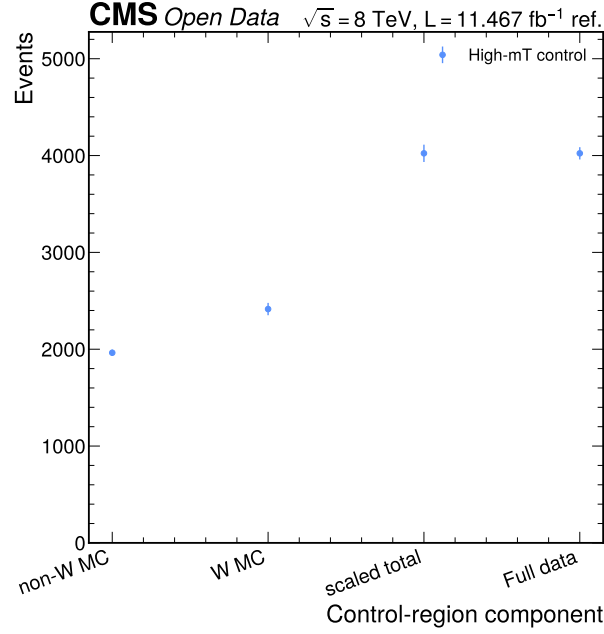


Figure 1: Full high-mT W control comparison. The figure shows non-W MC, nominal W MC, the scaled control-region prediction, and full data in the control region. The scale is derived outside the signal region and then propagated into the observed workspaces.

1.3 Primary Calibrated-Score Result

Category	Data	Background	QCD/fake	Data/background	Chi2/ndf	Max abs pull
vbf	79	60.17	0.36	1.313	1.513	2.36
boosted	2213	2115.53	5.64	1.046	1.904	2.63
zero_jet	8466	14047.12	22.93	0.603	4.025	3.22

The primary calibrated-score fit gives $\mu_{\text{hat}} = 38.3802$, an observed 95% CLs limit $\mu < 50.0000$, and a diagnostic q_0 value $Z = 12.4698$. The median expected limit from the same corrected workspace is 1.9735. This attempted optimized result is retained for comparison, but the final Phase 5 interpretation keeps the frozen visible-mass baseline because the optimized score fails the observed-data validation gate.

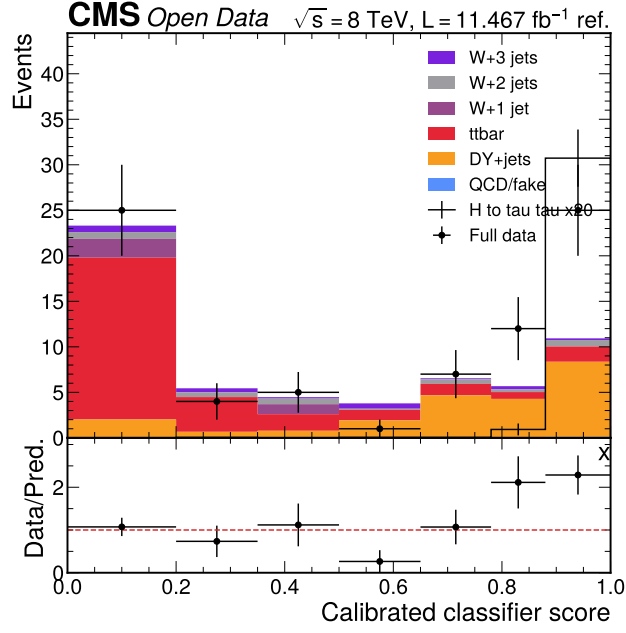


Figure 2: Primary calibrated-score validation in the VBF category. The plot compares localized Run2012B/C TauPlusX data to the calibrated score model after pre-training multivariate input reweighting, wider DY/Z normalization, and same-sign QCD/fake correction. This category remains statistically limited but is included in the simultaneous primary fit.

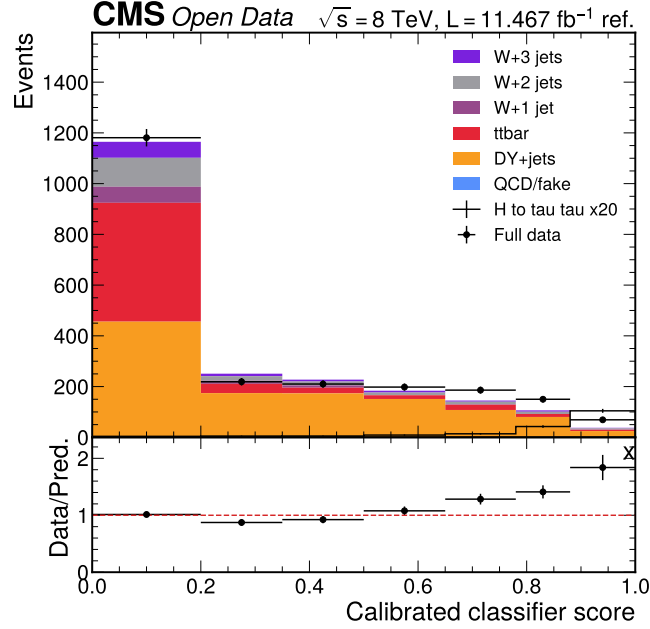


Figure 3: Primary calibrated-score validation in the boosted category. The plot compares full data to the calibrated score model with the W high-mT scale and same-sign QCD/fake template. This category is used in the final observed fit.

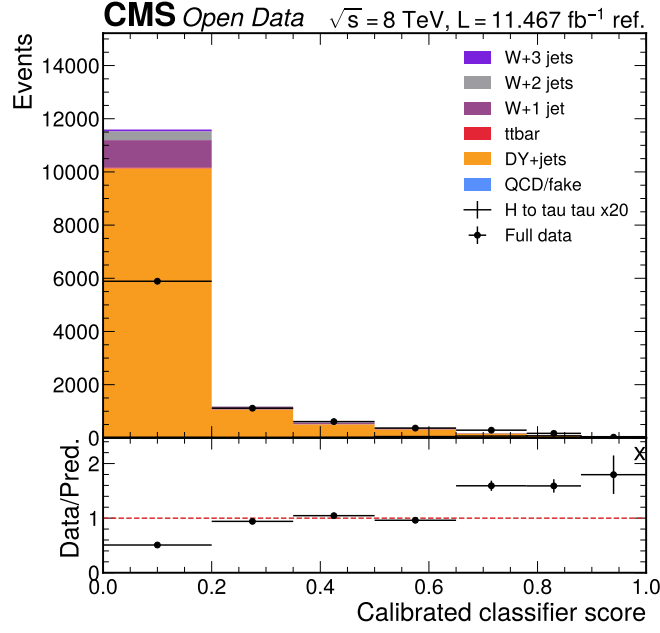


Figure 4: Primary calibrated-score validation in the zero-jet category. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate and the background input reweighting. This category contributes the largest data statistics to the primary simultaneous fit.

1.4 Frozen Baseline For Final Comparison

The frozen previous visible-mass baseline gives $\mu_{\text{hat}} = 0.4382$, observed 95% CLs $\mu < 10.7645$, median expected $\mu < 10.7375$. It is not retrained or refit in this update; it is carried forward only so the final AN and PRL can compare the optimized-score attempt with the previous result using the same signal-strength convention.

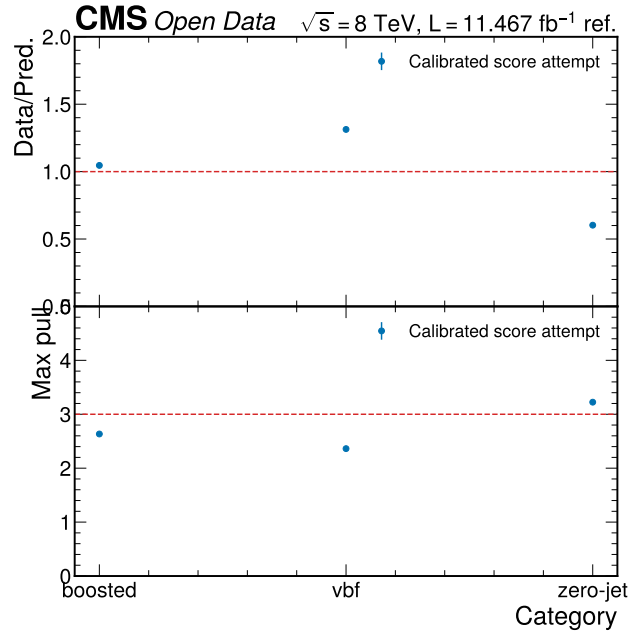


Figure 5: Observed pull and ratio summary. The figure compares the primary calibrated score validation behavior after the QCD/fake, DY/Z, and input reweighting corrections. It is the central diagnostic for the attempted optimized model.

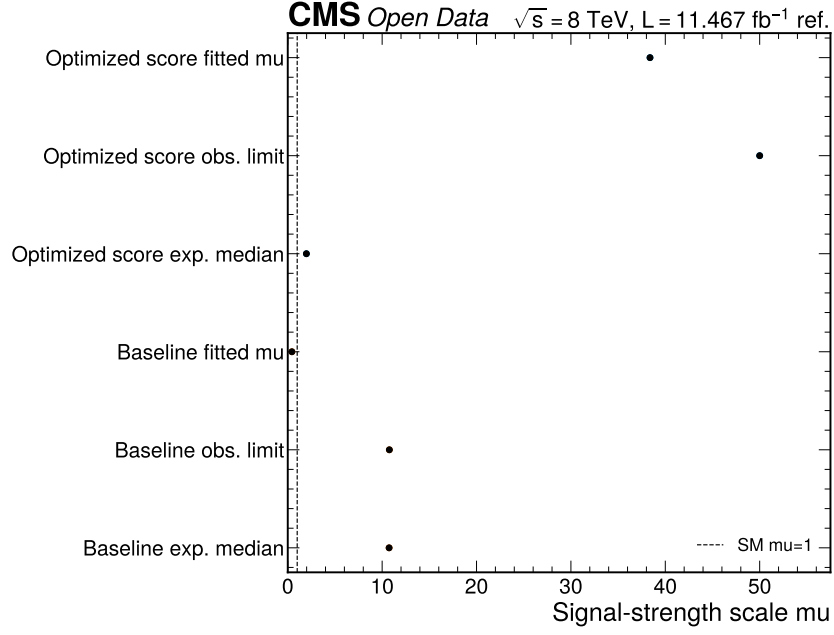


Figure 6: Observed limit and significance summary. The figure shows the primary calibrated-score fit and the frozen visible-mass baseline on the same signal-strength scale. The result remains an Open Data diagnostic rather than CMS-quality evidence.

1.5 Phase 5 Obligation

Phase 5 must report the calibrated-score plus QCD model as an optimized attempt, document the observed validation failure, and compare it with the frozen visible-mass baseline. It must also document the pre-training input reweighting, the widened DY/Z normalization, the absence of embedded/EWK Z reduced samples, and the 11.467/fb luminosity reference for these reduced tutorial samples.

References